

Digest Hardening Gates — Addendum Report

Date: 9 May 2026 | **Week:** 2026-W19 | **Scope:** 5 Monday-visible reps

Context: Root Cause of All Prior HOLD Decisions

Before addressing each of the six requested items, it is necessary to document a wiring bug discovered during this investigation. Every `repDigestGateResults` row written in week 2026-W19 recorded `decision = HOLD` for all five priority reps. The root cause was a field-stripping defect in `classifyBriefReadiness()` inside `emailDigest.ts`.

When constructing the `bestContact` object from the result of `selectProjectContact()`, the function only passed `{ name, title, email, linkedin }`. The `trustTier`, `source`, and `company` fields were silently dropped. The gate's `checkContactDefensibility()` function requires `trustTier === "send_ready"` and a non-null `source` to pass the `trust_tier_not_send_ready` and `card_detail_inconsistent` checks respectively. Because both fields arrived as `null`, every contact failed those two checks, producing a `card_detail_mismatch` blocker of severity `blocking`, which forced a HOLD regardless of the underlying contact quality.

The fix (applied in this session) adds `trustTier`, `source`, `company`, and `verificationScore` to the `bestContact` object at both the Monday and Thursday digest mapping sites, and extends the `DigestContact` interface to carry `source` and `verificationScore` through to the selector. TypeScript compilation confirms zero errors after the change.

1. Live Gate Replay — Before/After for All Five Reps

The simulation below uses the corrected gate logic against live DB data (active, non-suppressed projects with `send_ready` contacts). The “before” column reproduces the broken wiring by setting `trustTier = null` and `source = null`; the “after” column uses the actual DB values.

Rep	Business Line	Territory	Before (broken)	After (fixed)	Defensible/3
Ryan Pemberton	Portable Air	WA	HOLD	SEND	3/3
Brett Hansen	Pump/Dewatering	WA / NT	HOLD	SEND	3/3
Daniel Zec	Portable Air	NSW / VIC / SA / TAS	HOLD	SEND	3/3
Dan Day	Pump/Dewatering	SA / QLD / VIC / NSW / TAS	HOLD	SEND	3/3
Amit Bhargava	PAL	National	HOLD	SEND	3/3

All five reps achieve a clean 3/3 defensible-contact result once the wiring is corrected. The table below shows the actual top-3 projects and primary contacts that would appear in each digest.

Ryan Pemberton — WA, Portable Air

Project	Score	State	Primary Contact	Title	Email
Murchison Gold Project underground development	100	WA	Win Comia	Underground Manager	wcomia@meekametals.com.au
United North Underground Gold Mine	100	WA	Jonathan Streeter	General Manager: Geology	jstreeter@roxresources.com.au
Walyerling West-1 Gas Development	99	WA	Tom Luke	Project Director-Project Haber	tom.luke@strikeenergy.com.au

Brett Hansen — WA / NT, Pump/Dewatering

Project	Score	State	Primary Contact	Title	Email
Alkimos Desalination Plant	100	WA	Rob Gordon	Procurement Manager, Acciona	rob.gordon@acciona.com.au
Pluto LNG Facility Operations	100	WA	Paul Moscardini	Site Manager	paul.moscardini@woodside.com.au
Kwinana Gas Power Generation 2 Project	100	WA	Johan Myburgh	General Manager Construction	jmyburgh@agl.com.au

Daniel Zec — NSW / VIC / SA / TAS, Portable Air

Project	Score	State	Primary Contact	Title	Email
Snowy 2.0 Hydroelectric Project	100	NSW	Giuseppe Gulisano	Deputy Construction Manager	g.gulisano@futuregenerationjv.com.au
Bass Strait Decommissioning Infrastructure Upgrade	100	VIC	Darren Whyte	Fleet Manager	darren.whyte@qube.com.au
Tunkillia Gold Project	99	SA	Steven Donhardt	Project Manager, Ausdrill	steven.donhardt@ausdrill.com.au

Dan Day — SA / QLD / VIC / NSW / TAS, Pump/Dewatering

Project	Score	State	Primary Contact	Title	Email
Warrnambool Sewage Treatment Plant Upgrade	100	VIC	Ahsan Khan CPEng	Senior Mechanical Engineer, Downer	ahsan.intpe@downer.com.au
Victorian Water Sector Construction Activity Increase	100	VIC	Eric Pritchard	Project Manager, John Holland	eric.pritchard@johnholland.com.au
Bruce Highway Targeted Safety Program (BHTSP)	100	QLD	Marc Regonesi	Procurement Manager, CPB Contractors	marc.regonesi@cpbcontractors.com.au

Amit Bhargava — National, PAL

Project	Score	State	Primary Contact	Title	Email
Regional Road Upgrade	92	NT	Trevor Hopps	Operations Centre Systems Manager	trevor.hopps@nt.gov.au
Fortescue 4-5GWh BESS with 1.8GW Renewable Energy	92	—	Wayne Mills	Construction Manager	wayne.mills@fortescue.com
Bruce Highway Safety Upgrades Program	91	QLD	Rick McConnon	Manager Business - Procurement	rick.mcconnon@queenslanddepartmentoftransportandmainroads.com.au

2. Card / Detail Consistency

The `selectProjectContact()` function in `contactSelector.ts` is the **single source of truth** for contact selection. It is called identically from:

- `classifyBriefReadiness()` in `emailDigest.ts` — which produces the `bestContact` shown on the email summary card
- `getThisWeekForEmail()` in `thisweekService.ts` — which produces the contact shown on the “This Week” detail section
- The project detail page API via `trpc.projects.getById` — which returns all `send_ready` contacts sorted by `verificationScore` descending

The card always shows the contact that `selectProjectContact()` returns as `selectedContact`. The detail page shows the same contact first (highest `verificationScore`) followed by any additional `send_ready` contacts for that project. There is no separate selection path that could diverge.

The following table verifies this for the top WA Portable Air projects (the highest-traffic rep territory):

Project	Card Contact	Verification Score	Additional Send-Ready Contacts
Murchison Gold Project underground development	John Sinagra	95	Win Comia, Matthew O'Hara
United North Underground Gold Mine	Jonathan Streeter	95	Oliver Keene
Walayering West-1 Gas Development	Diego Linares	95	Bas Voorbrood, Allan Rogers, Andrew Farley (+6 more)
Mount Ridley Heavy Rare Earth Resource	Allister Caird	92	—

Result: PASS on all checked projects. The card contact and the detail page primary contact are identical because both draw from the same sorted pool via the same function.

3. Operator Review Path

What a Stored `repDigestGateResults` Row Looks Like

The following is the actual structure of a row written by `storeGateResult()` for week 2026-W19 (Kevin Arnandes, representative example):

```
{
  "id": 8,
  "userId": <user_id>,
  "weekKey": "2026W19",
  "phase": "pre_digest",
  "decision": "HOLD",
  "rescueAttempted": false,
  "top3Snapshot": [
    { "id": 690037, "name": "Fortescue 4-5GWh Battery Storage...", "score": 88, "contactName": "Clarence Titus" },
    { "id": 690030, "name": "Fortescue 4-5GWh BESS with 1.8GW...", "score": 88, "contactName": "Wayne Mills" },
    { "id": 660008, "name": "Queensland Copper Mine Off-grid...", "score": 88, "contactName": "..." }
  ],
  "blockers": [
    {
      "criterion": "contact_not_defensible",
      "detail": "\"Fortescue 4-5GWh Battery Storage...\" contact \"Clarence Titus\" failed: trust_tier_not_send_ready, card_detail_inconsistent",
      "severity": "warning"
    },
    {
      "criterion": "card_detail_mismatch",
      "detail": "\"Fortescue 4-5GWh Battery Storage...\" contact \"Clarence Titus\" has inconsistent card/detail data",
      "severity": "blocking"
    }
  ],
  "createdAt": "2026-05-09T03:39:24.000Z"
}
```

How SEND vs HOLD Is Reviewed in Practice

The operator review path works as follows:

- 1. Automated logging.** Every Monday digest run writes one `repDigestGateResults` row per rep before attempting to send. The `phase` field is `"pre_digest"`. If a post-pipeline snapshot was taken, a separate row with `phase = "post_pipeline"` also exists.
- 2. Blocker inspection.** Each blocker has a `severity` field. Only `"blocking"` severity blockers prevent the send. `"warning"` severity blockers are logged for visibility but do not stop the digest. The operator can query `SELECT * FROM repDigestGateResults WHERE decision = 'HOLD' AND weekKey = '2026W19'` to see all held reps and their specific blockers.
- 3. Manual override path.** There is currently no UI override button. The operator must either (a) fix the underlying data issue (promote the contact's `contactTrustTier`, correct the `source` field) and re-run the digest, or (b) temporarily set `phase = "manual"` and insert a `decision = "SEND"` row to bypass the gate for that rep.

4. **Email log cross-reference.** The `userEmailSendLog` table records `status = "failed"` with `error = "REP_HARDENING_GATE_HELD: ..."` for each held rep, providing a second audit trail visible in the Email Logs UI panel.

4. Automatic Narrow Rescue Trigger

Is It Automatic?

The `identifyRescueCandidates()` function in `digestHardeningGates.ts` is **defined and exported** but is **not yet called automatically** from the digest pipeline. The current `emailDigest.ts` call to `runAllGates()` does not invoke `identifyRescueCandidates()`, and `rescueAttempted` is always stored as `false`. The rescue trigger is implemented as a standalone function that must be called explicitly — it is not wired into the Monday run.

Rescue Logic (as Implemented)

When called, `identifyRescueCandidates()` applies the following rules:

Rule	Value
Scope	Top 5 visible projects per rep only
Minimum relevance score	≥ 40
Lane fit required	High or Medium
Skip if already has ≥ 2 <code>send_ready</code> contacts	Yes
Cooldown window	7 days since last enrichment
Max rescues per run	3 projects
Apollo budget reserve	5 credits held back for post-delta pass
Effective daily budget	<code>apolloDailyCap - apolloDailyUsed - 5</code>

Budget Guardrails

Apollo enrichment is governed by a hard daily cap of **200 credits** enforced in `apolloEnrichment.ts` (`DAILY_ENRICHMENT_CAP = 200`). The `wasRecentlyRevealed()` dedup guard checks `apolloCreditLog` for any reveal of the same `apolloPersonId` within a 168-hour (7-day) window and skips the call if found. A second guard in `revealContactEmail()` checks the contact's `enrichmentStatus` and `enrichedAt` timestamp, refusing to re-reveal a contact enriched within the past 7 days. Current usage is 16 credits today against the 200-credit cap.

Provider Dedup Protections

Three layers prevent duplicate Apollo spend:

1. **Person-level dedup:** `wasRecentlyRevealed(apolloPersonId, 168)` — skips if the same Apollo person was revealed in the last 7 days.
2. **Contact-level dedup:** `revealContactEmail()` checks `enrichmentStatus = "enriched"` within 7 days before calling the API.
3. **Project-level cache:** `getProjectEnrichmentCache()` returns a cache hit if the project was enriched within the cache window, preventing re-enrichment of the same project's company roster.

Gap: Rescue Is Not Yet Wired

The rescue trigger must be connected to the Monday pipeline. The recommended wiring is to call `identifyRescueCandidates()` after the gate returns `HOLD`, pass the result to `storeGateResult()` with `rescueAttempted: true`, and then invoke the Apollo enrichment pipeline for the identified candidates before re-running the gate. This is a known pending item.

5. Overlap / Differentiation Check

Project-Level Overlap

The table below shows whether any project appears in two reps' top-5 lists simultaneously.

Pair	Overlap	Notes
Ryan (WA, PA) vs Brett (WA/NT, Pump)	0	Different business lines; territory overlap does not produce shared projects
Ryan (WA, PA) vs Daniel (NSW/VIC/SA/TAS, PA)	0	Fully differentiated by territory
Ryan (WA, PA) vs Dan Day (SA/QLD/VIC/NSW/TAS, Pump)	0	Different BL and territory
Ryan (WA, PA) vs Amit (National, PAL)	0	Different BL
Brett (WA/NT, Pump) vs Daniel (NSW/VIC/SA/TAS, PA)	0	Different BL and territory
Brett (WA/NT, Pump) vs Dan Day (SA/QLD/VIC/NSW/TAS, Pump)	0	Territory split is clean
Brett (WA/NT, Pump) vs Amit (National, PAL)	2	Projects 570009 (Regional Road Upgrade, NT) and 570001 (NT Flood Damaged Roads) appear in both
Daniel (NSW/VIC/SA/TAS, PA) vs Dan Day (SA/QLD/VIC/NSW/TAS, Pump)	0	Different BL
Daniel (NSW/VIC/SA/TAS, PA) vs Amit (National, PAL)	0	Different BL
Dan Day (SA/QLD/VIC/NSW/TAS, Pump) vs Amit (National, PAL)	0	Different BL

Do the Gates Improve Differentiation?

The gates do not change which projects appear in each rep's list — that is determined by `scoreAndFilterProjects()` and territory filtering upstream. What the gates do is prevent a rep from receiving a digest that contains a project with a defensible-but-wrong contact (e.g., a fabricated email, a non-industrial title, or an LLM-inferred address). This is a **safety improvement**, not a differentiation improvement.

The two-project overlap between Brett and Amit (NT infrastructure projects) is a genuine gap. Both reps see the same NT road projects because Brett's territory includes NT and Amit is national. The gate does not resolve this; it would require either a territory exclusivity rule or a business-line exclusivity rule at the digest assembly stage. This is a known open item.

Ryan vs Brett

Ryan receives WA Portable Air projects (mining, underground, gas). Brett receives WA/NT Pump/Dewatering projects (desalination, LNG, power generation). The top-3 lists are completely non-overlapping. The gate confirms both would SEND with $\frac{3}{3}$ defensible contacts. No differentiation concern.

Daniel vs Dan Day vs Amit

Daniel receives NSW/VIC/SA/TAS Portable Air projects (tunnelling, rail, hydroelectric, mining). Dan Day receives SA/QLD/VIC/NSW/TAS Pump/Dewatering projects (water treatment, highway, solar-storage). Amit receives national PAL projects (pumped hydro, BESS, road upgrades). The only overlap is Brett/Amit on NT road projects as noted above. Daniel and Dan Day share territory (SA, VIC, NSW, TAS) but different business lines, producing zero project overlap. All three would SEND with $\frac{3}{3}$ defensible contacts.

6. Credit Safety

Will the Hardening Path Recreate Duplicate Apollo Waste?

The answer is **no**, provided the rescue trigger is wired correctly. The following protections are already in place and confirmed active:

Protection	Mechanism	Status
Person-level dedup	<code>wasRecentlyRevealed(apolloPersonId, 168h)</code> in <code>apolloEnrichment.ts</code>	Active
Contact-level dedup	<code>enrichmentStatus = "enriched" + enrichedAt < 7 days</code> check in <code>revealContactEmail()</code>	Active
Project-level cache	<code>getProjectEnrichmentCache()</code> prevents re-enrichment of same project within cache window	Active
Daily hard cap	<code>DAILY_ENRICHMENT_CAP = 200</code> in <code>apolloEnrichment.ts</code>	Active
Rescue budget reserve	<code>BUDGET_RESERVE = 5</code> credits held back in <code>identifyRescueCandidates()</code>	Implemented, not yet wired
Rescue cooldown	7-day cooldown per project in <code>identifyRescueCandidates()</code>	Implemented, not yet wired
Max rescue per run	3 projects per run in <code>identifyRescueCandidates()</code>	Implemented, not yet wired

Current Apollo usage is **16 credits today** (512 reveal credits over the past 7 days), well within the 200/day cap. The wiring bug that caused all five reps to HOLD did not trigger any rescue attempts (`rescueAttempted = false` on all rows), so no duplicate spend occurred from the gate failure itself.

The one credit-safety risk that remains is if the rescue trigger is wired without the cooldown check, which would allow the same project to be re-enriched on every Monday run. The `identifyRescueCandidates()` function already contains the 7-day cooldown guard; the implementation requirement is simply to pass `project.lastEnrichedAt` correctly when calling it.

Summary of Open Items

Item	Status	Priority
Fix <code>bestContact</code> wiring bug (trustTier, source, company not passed)	Fixed in this session	Done
Wire <code>identifyRescueCandidates()</code> into Monday pipeline	Not yet wired	High
Resolve Brett / Amit NT project overlap (2 shared projects)	Known gap, no fix yet	Medium
Add operator UI override for HOLD decisions	Not implemented	Medium
Confirm <code>selectProjectContact()</code> is called on detail page API (not a separate sort)	Confirmed consistent	Done